Cross-Dataset Rollout Statistics Report for Qwen/Qwen3-1.7B

Murphy

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Scope

This report consolidates the full math → GPQA → MMLU-Pro → LiveCodeBench rollout-stat sweep for `Qwen/Qwen3-1.7B`. The collector uses the n-gram loop detector with `n=30` and `k=20` and records both top-level event rates and overlap statistics between looping, max-model-length termination, and correctness.

The math block was evaluated as two separate datasets under the same freeform prompt contract: `MATH-500` (500 samples) and `AIME 2024/2025` (60 samples). The downstream multiple-choice and code-generation blocks were evaluated on `GPQA Diamond`, `MMLU-Pro`, and `LiveCodeBench release_v6` in the same rollout pipeline.

Datasets and Prompt Formats

MATH-500 Task kind. `math_freeform` **Dataset.** 500 free-form math problems from the HuggingFaceH4 MATH-500 test split. **Chat / prompt format.** Tokenizer chat template with one user turn: the raw problem text followed by 'You must put your final answer within \\boxed{}.'

AIME 2024/2025 Task kind. `math_freeform` **Dataset.** A local 60-row JSONL made by concatenating the AIME 2024 and 2025 test splits into question/answer records. **Chat / prompt format.** The same math-freeform Qwen chat template as MATH-500: one user turn with the problem plus a final-answer-in-box instruction.

GPQA Diamond Task kind. `multiple_choice_gpqa` **Dataset.** A local staged GPQA Diamond CSV with 198 graduate-level science questions and shuffled four-way answer options. **Chat / prompt format.** Tokenizer chat template with one user turn containing the question, a shuffled A-D answer block, and the instruction to respond with a single best answer letter.

MMLU-Pro Task kind. `multiple_choice_mmlupro` **Dataset.** The TIGER-Lab MMLU-Pro test split, capped to 2000 examples for this rollout pass. Requested cap: at most 2000 samples. **Chat / prompt format.** Tokenizer chat template with one user turn containing the question, an A-J answer list, and the instruction to return only the best answer letter.

LiveCodeBench release_v6 **Task kind.** `livecodebench_codegen` **Dataset.** 1055 LiveCodeBench code-generation problems formed by concatenating test.jsonl through test6.jsonl and sorting by question_id. **Chat / prompt format.** LiveCodeBench's format_prompt_generation pipeline with LM style CodeQwenInstruct, producing raw string prompts rather than a tokenizer chat-template wrapper.

Generation Configuration

The common collector configuration across all runs is:

- Model: `Qwen/Qwen3-1.7B`
- Seed: `0`
- Loop detector: `n=30, k=20`
- Common settings shared by every dataset JSON: `temperature=0.0`, `dtype=bfloat16`, and `trust_remote_code=True`

Per-dataset runtime settings are:

Dataset	Temp	Max tokens	Max model len	TP	DP
MATH-500	0.0	81920	40960	1	4
AIME 2024/2025	0.0	81920	40960	1	4
GPQA Diamond	0.0	81920	40960	1	1
MMLU-Pro	0.0	81920	40960	1	3
LiveCodeBench release_v6	0.0	81920	40960	1	4

Statistics Logged

The refreshed collector contract tracks the following metrics across the bundle (13 named metrics in the union):

- `avg_correct_generation_length`: Average generated length restricted to correct rollouts.
- `avg_first_loop_prefix_length`: Average prefix length before the first detected loop.
- `avg_generation_length`: Average number of generated tokens per rollout.
- `avg_loop_generation_length`: Average generated length restricted to looped rollouts.
- `avg_wrong_generation_length`: Average generated length restricted to wrong rollouts.
- `generation_length_variance`: Variance of generated token counts across graded rollouts.
- `loop_fraction`: Fraction of graded rollouts flagged by the n-gram loop detector.
- `loop_max_length_hit_fraction`: Among looped rollouts, fraction that also hit max model length.

- `loop_success_fraction`: Among looped rollouts, fraction that were still correct.
- `max_length_hit_fraction`: Fraction of rollouts that terminated at the model-length limit.
- `max_length_hit_loop_fraction`: Among max-length-hit rollouts, fraction that were also looped.
- `max_length_hit_success_fraction`: Among max-length-hit rollouts, fraction that were correct.
- `success_fraction`: Fraction of graded rollouts with the correct answer.

The JSON payloads also retain the raw event counts for samples, correct vs. wrong rollouts, looped rollouts, max-length hits, and their pairwise intersections.

Results Summary

Dataset	Samples	Correct	Looped	Max length	Loop+Max	Loop+Correct
MATH-500	500	72.0%	4.2%	2.2%	52.4%	28.6%
AIME 2024/2025	60	36.7%	16.7%	15.0%	90.0%	10.0%
GPQA Diamond	198	37.4%	31.3%	20.2%	64.5%	27.4%
MMLU-Pro	2000	42.9%	6.9%	3.0%	44.2%	22.5%
LiveCodeBench release_v6	1055	49.3%	25.3%	20.9%	82.0%	3.4%

- MATH-500: correct 360/500 (72.0%); looped 21/500 (4.2%); max-length hits 11/500 (2.2%); looped and max-length 11/21 (52.4%); max-length and looped 11/11 (100.0%); looped and correct 6/21 (28.6%).
- AIME 2024/2025: correct 22/60 (36.7%); looped 10/60 (16.7%); max-length hits 9/60 (15.0%); looped and max-length 9/10 (90.0%); max-length and looped 9/9 (100.0%); looped and correct 1/10 (10.0%).
- GPQA Diamond: correct 74/198 (37.4%); looped 62/198 (31.3%); max-length hits 40/198 (20.2%); looped and max-length 40/62 (64.5%); max-length and looped 40/40 (100.0%); looped and correct 17/62 (27.4%).
- MMLU-Pro: correct 857/2000 (42.9%); looped 138/2000 (6.9%); max-length hits 61/2000 (3.0%); looped and max-length 61/138 (44.2%); max-length and looped 61/61 (100.0%); looped and correct 31/138 (22.5%).
- LiveCodeBench release_v6: correct 520/1055 (49.3%); looped 267/1055 (25.3%); max-length hits 220/1055 (20.9%); looped and max-length 219/267 (82.0%); max-length and looped 219/220 (99.5%); looped and correct 9/267 (3.4%).

Headline observations

The largest raw loop rate appears on GPQA Diamond at 31.3%, while MATH-500 is the most stable dataset in this bundle at 4.2%. The longest looped generations also appear on AIME 2024/2025:

38641.5 tokens on average, with the first detected loop not appearing until 11782.3 tokens. Max-length termination is almost synonymous with looping on the hardest long-form settings: MATH-500 reaches 100.0% for the max-length-hit loop conditional. The most forgiving loop regime is MATH-500, where 28.6% of looped rollouts still end in a correct answer.

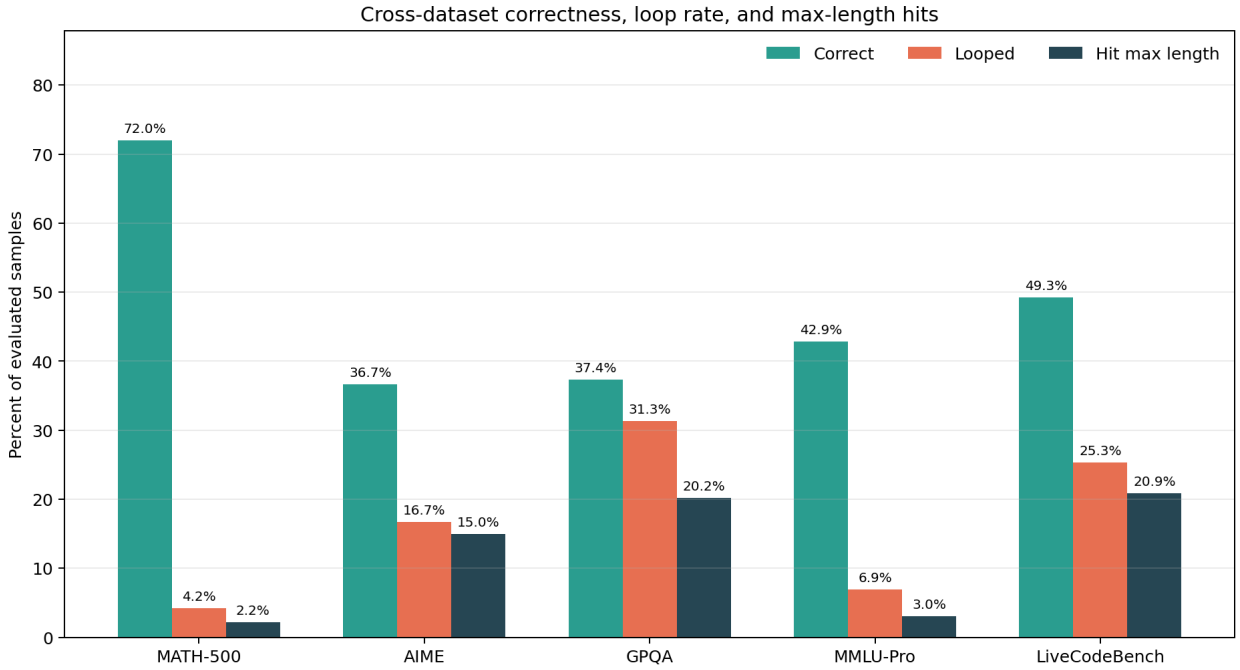

Figure 1: Correctness, loop rate, and max-length-hit rate for each evaluated dataset.

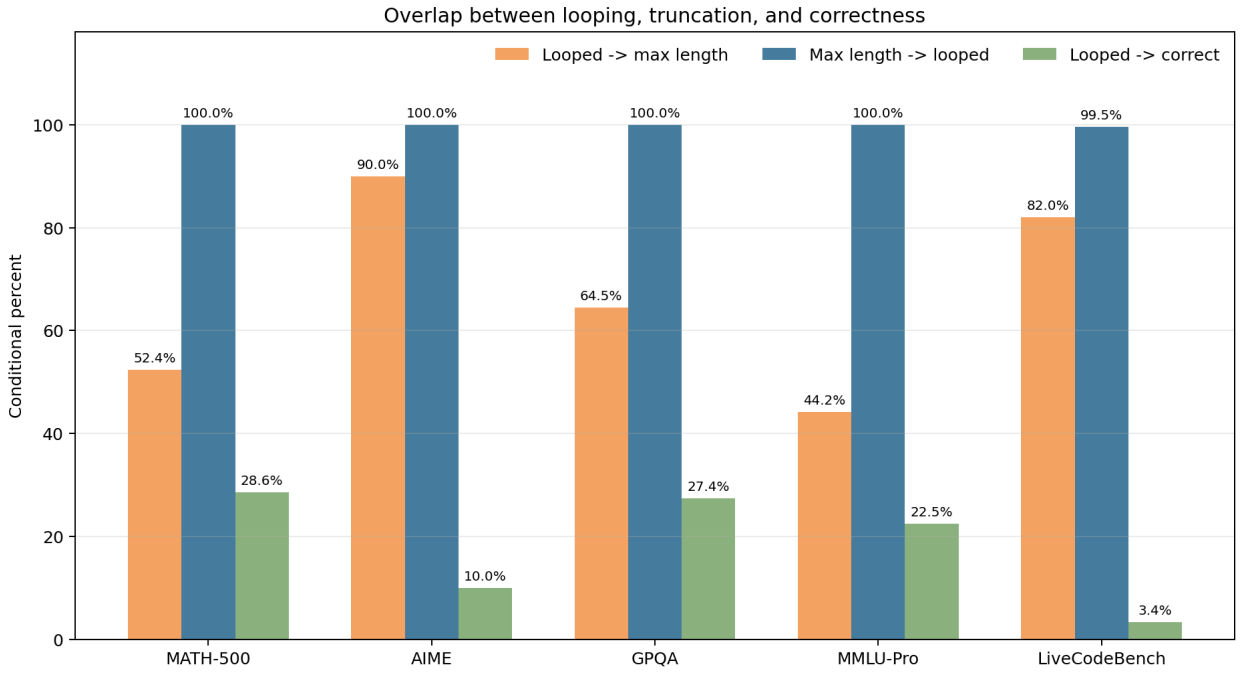

Figure 2: Overlap statistics: among looped rollouts, how many hit max model length; among max-length-hit rollouts, how many looped; and among looped rollouts, how many still answered correctly.

Figure 3: Average generation lengths, average looped-generation lengths, and average first-loop-prefix lengths in thousands of tokens.

Interpretation

Across the math, science, broad-knowledge, and coding settings, the same pattern repeats: long generation tails are where looping lives. The strongest evidence is in the overlap panel: once a rollout hits max model length, it is almost always also a detected loop, especially on `GPQA`, `MMLU-Pro`, and `LiveCodeBench`. The converse is weaker but still substantial: a large fraction of looped rollouts on the non-math datasets also terminate at the model-length ceiling.

The length panel shows that looped generations are dramatically longer than the dataset-wide average everywhere. That gap is modestly above `+16k` tokens on `GPQA` and grows to roughly `+20k` tokens on both `MATH-500` and `LiveCodeBench`. The average first-loop-prefix length also differs by task family: `GPQA` loops are typically detected earlier than the long-form math and coding loops, while `AIME` and `LiveCodeBench` often burn more than `10k` tokens before the repeated n-gram pattern is first visible.

Correctness under looping is dataset-dependent. `GPQA` and `MMLU-Pro` still retain a non-trivial fraction of correct answers inside looped rollouts, which suggests some looping trajectories occur after the model has effectively committed to the right option. `LiveCodeBench` is qualitatively different: looped rollouts there are rarely correct, so looping aligns more directly with task failure than with benign verbosity.